

1 Introduction



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03 Real Analysis I

Instructor: David Earn

Lecture 1
Introduction
Tuesday 6 January 2026

If you haven't already, please fill in Math 3A03 Survey 1 on childsmath

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Survey 1**
- .

Where to find course information

- The course web site: <http://ms.mcmaster.ca/earn/3A03>
- Click on [Course syllabus](#) to view the basic course information.
 - *You are expected to read and pay attention to every word of the syllabus.*
 - *Note that some items in the syllabus are tentative and might be updated.*
- Let's have a look now. . .

What you are assumed to know

Chapters 1–5 of Bartle and Sherbert (BS), *Introduction to Real Analysis*

- The Real Numbers, $\mathbb{R}$ (BS Chapter 2)
- Sequences and Series (BS Chapter 3)
- Limits of functions (BS Chapter 4)
- Continuous functions (BS Chapter 5)

Let's do a few polls to trigger your memory. . .

Background Poll: Characteristic property of $\mathbb{R}$

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Background: The property that characterizes $\mathbb{R}$ in comparison with $\mathbb{Q}$**
- .

Background Poll: Sequence limit order

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Background: Sequence limit order**
- .

Background Poll: Monotone convergence

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Background: Monotone convergence**
- .

Background Poll: Intermediate and extreme values

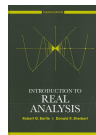
- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Background: Intermediate and extreme values**
- .

What we will cover in this course

Primary sources

■ *Second half of Math 3IA3 textbook:*

Bartle and Sherbert (BS),
Introduction to Real Analysis



■ *Online textbook:*

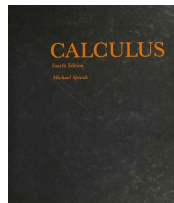
Thomson, Bruckner and Bruckner (TBB),
Elementary Real Analysis



■ *Optional book:*

Michael Spivak, "Calculus" (4th ed., 2008)

- This book is not sufficiently advanced for this course, but the exposition of elementary analysis is very nice.
- It is a great read, and if you read it you'll see some of the things that inspire some aspects of my presentation style.



What we will cover in this course

Primary topics

- ***Differentiation***

(BS Chapter 6) (TBB Chapter 7)

- ***Integration***

(BS Chapter 7) (TBB Chapter 8)

- ***Topology of the real line***

(BS Chapter 11) (TBB Chapter 4)

- ***Sequences and series of functions***

(BS Chapters 8 and 9) (TBB Chapter 9)

- ***Metric spaces***

(TBB Chapter 13)

Survey

If you haven't already,
please complete the "Survey 1"
poll on childsmath.

In addition to doing the survey, please e-mail
me (now or at any point in the term) with
any preferences or suggestions you have
about how to run the course.

Questions

Questions?